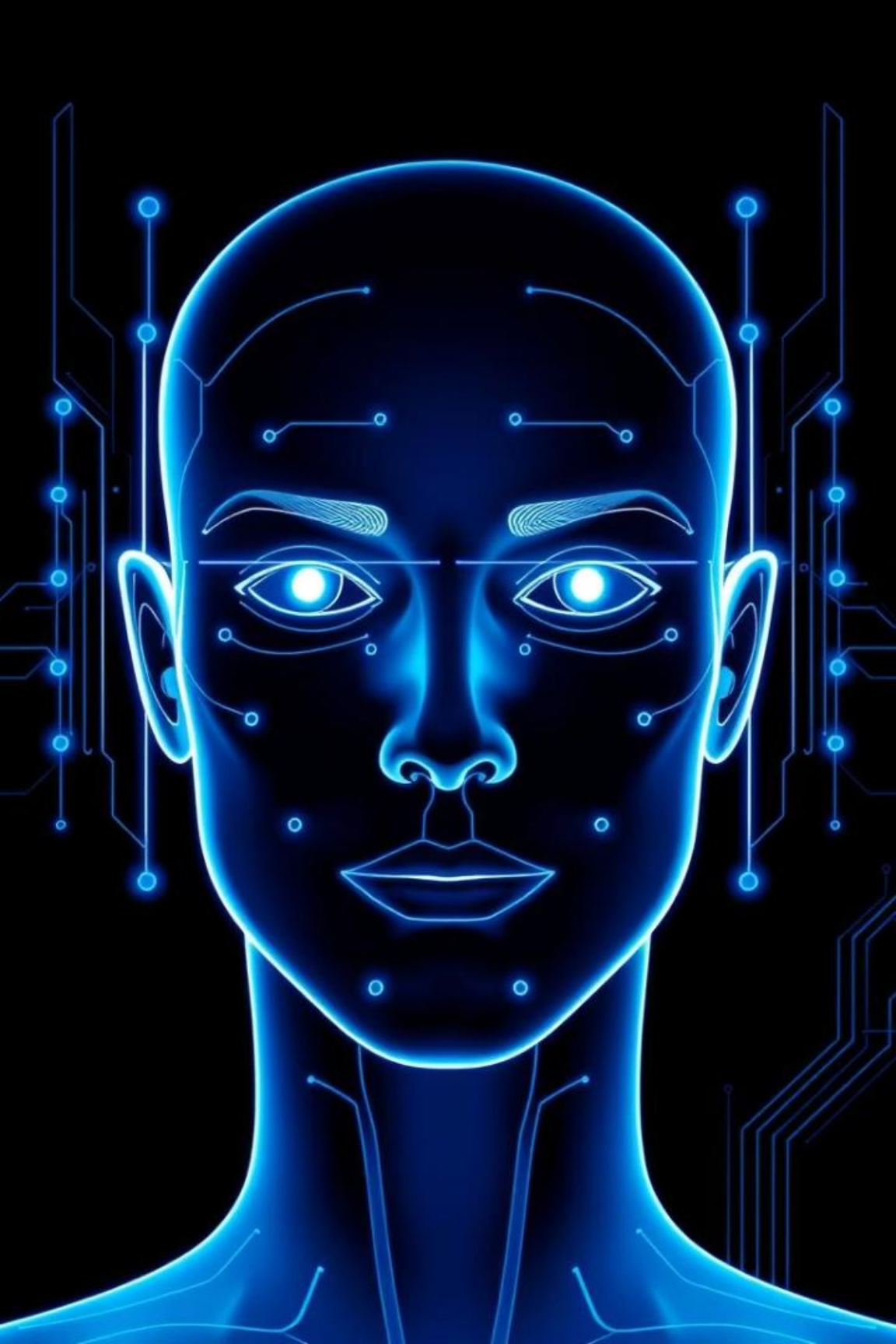




Face Recognition Attendance System

Automating Attendance Tracking with AI

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Problem Statement

Manual System Issues

- Time-consuming roll calls delay classes
- Proxy attendance by friends
- Human errors and lost records

Our AI Solution

- AI-based facial recognition
- Zero physical contact, post-pandemic safe

Key Objectives

Automate Attendance

Real-time face detection replaces paper

High Accuracy

Over 85% recognition rate with LBPH

Prevent Fraud

Biometric verification stops proxy attendance

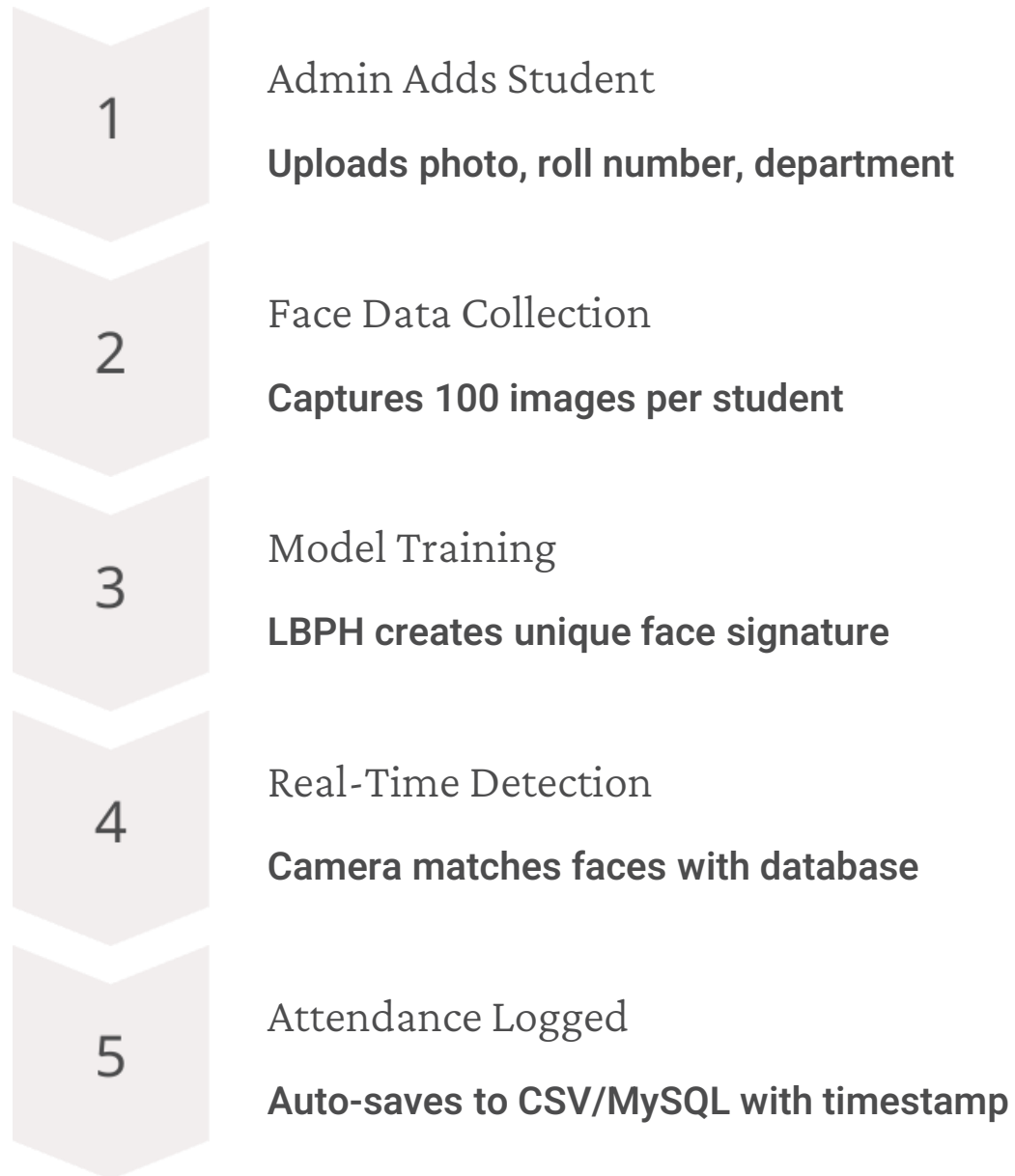
User-Friendly Interface

Simple GUI to add students and view logs

Scalability

Supports 1000+ students with MySQL

System Workflow



Technical Stack

Face Detection

OpenCV Haar Cascade

Face Recognition

**LBPH Algorithm:
lightweight, accurate**

Database

**MySQL stores roll numbers,
departments**

GUI

Tkinter, Python-compatible

Backend

Python + OpenCV

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Code Highlights

Face Detection

detectMultiScale for real-time face detection

Attendance Logging

Append attendance with timestamp to CSV

Database Integration

Insert student data into MySQL

Dataset & Training

Dataset Structure

Folder: data/ with user.{rollno}.{id}.jpg files

Training Process

- Convert images to grayscale
- Extract LBPH features
- Save model as classifier.xml

Results & Performance

- Accuracy: 85-90% in good lighting
- Speed: 10-15 faces/second at 720p
- Storage: MySQL handles 10,000+ records
- CSV for daily backups

Metric	Traditional	Our System
Time/Student	30 sec	2 sec
Proxy Risk	High	None

Challenges & Fixes

Challenge	Solution
Low-light recognition	Adaptive thresholding added
False positives	Increased training samples
Database lag	Batch processing implemented



Future Scope

- 1 — 2024: Mobile App
Android/iOS attendance marking
- 2 — 2025: Cloud Sync
Live logs on AWS/Azure
- 3 — Deep Learning Upgrade
CNN for 95%+ accuracy
- 4 — Multi-Factor Auth
Face + RFID/NFC

